

Ionic bonding

Introduction

Ionic bond: the **strong electrostatic attraction** between **oppositely charged ions** formed when a metal reacts with a non-metal.

In an ionic compound the ions are held in a repeating pattern (alternating +ive and -ive) called a **giant ionic lattice** structure where the attraction force between opposite charges is maximized while repulsion between similar ones is reduced.

note: ion attraction forces do not rely on orientation nor direction (omnidirectional)

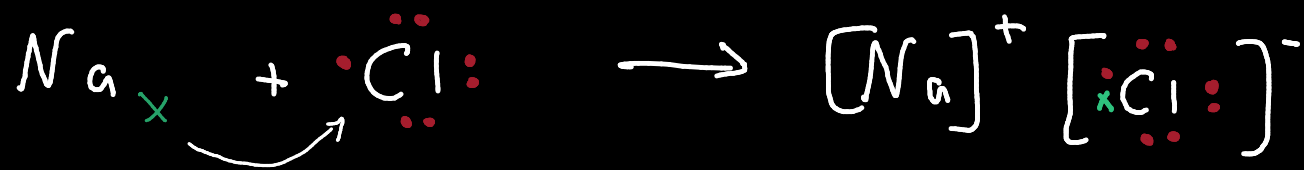
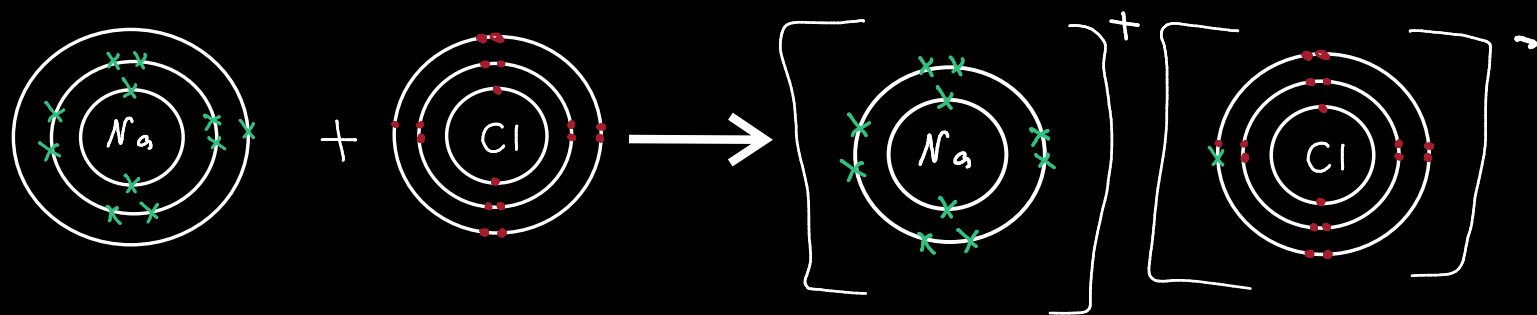
Each ion is trying to form a full outer shell (octet rule) and this is achieved by:

Metals: losing an electron forming positive ions (cations)

Non-metals: gaining an electron forming a negative ion (anion)

Each ion is isoelectronic to the nearest noble gas

Isoelectronic: Particles (atoms, ions, or molecules) that have the same number of electrons and the same **electron configuration**.



Evidence of existence of ions

Physical properties

1. High Melting and Boiling Points

Explanation: Ions are held by strong electrostatic forces in a giant lattice. A large amount of energy is required to overcome these forces.

Result: This results in a high melting point.

2. Brittleness

General Rule: Ionic compounds are considered to be brittle.

Mechanism: When a force is applied, it can cause layers of ions to shift. This movement places ions of the same charge next to each other, which repel (like charges repel), causing the material to shatter.

3. Electrical Conductivity

Solid State: Not conductive.

—Reason: Ions are fixed in place and there are no delocalised electrons.

Molten or Aqueous Solution: Conductive.

—Reason: The ions are free to move.

Electrolysis: If a current is applied, the compound will undergo electrolysis

Cations → negative electrode (cathode) and are reduced.

Anions → positive electrode (anode) and are oxidized.

4. Solubility

General Rule: Ionic compounds are soluble in water.

Mechanism: The bond breaking is caused by interaction (attraction) between ions and water molecules followed by hydration.

Hydration: The process where water molecules are attracted to ions and surround them due to water's polarity (will be explained in further details in covalent bonds)

—Oxygen end (slightly negative) attracted to positive ion.

—Hydrogen end (slightly positive) attracted to negative ion.

—This forms an ion-dipole bond.

Migration of ions in electrolysis

Like what's mentioned earlier, ionic compounds undergo electrolysis when direct current is passed through it (in its molten or aqueous states)

During this process:

—Positive ions (e.g. Na^+) migrate towards the negative electrode where they gain electrons forming atoms (e.g. Na atoms).

—Negative ions (e.g. Cl^-) migrate towards the positive electrode where they lose electrons forming atoms/molecules (e.g. Cl atoms).

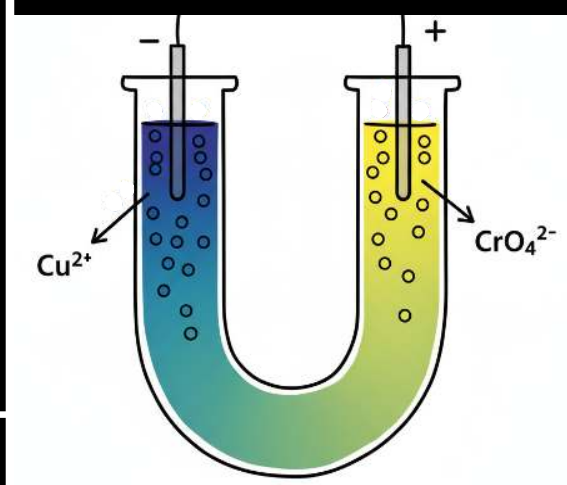
This concept could be demonstrated even further using colored ions such as:

—Copper ion (Cu^{2+}) which has a **blue color**

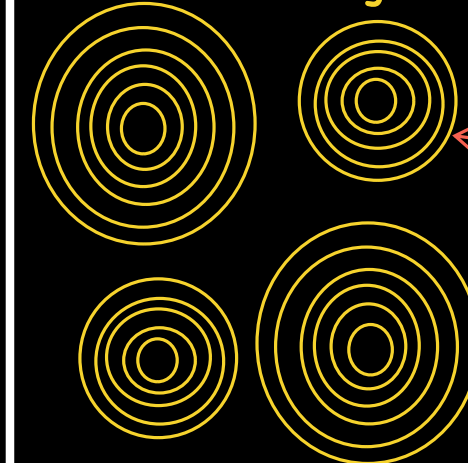
—Chromate ion (CrO_4^{2-}) which has a **yellow color**

—Manganate ion (MnO_4^-) which has a **purple**

(During electrolysis, the area around each electrode will turn the color of the ion migrating towards it)



Electron density maps



Observation: Empty space between ions

Explanation: Electron density is zero/low which means electrons are not shared (unlike covalent bonds). This indicates that this is an ionic compound

Strength of ionic bonding

Lattice Enthalpy (energy): The enthalpy change when 1 mole of ionic compound is formed from its **gaseous ions** in standard conditions. (exothermic)

A **more** Negative enthalpy → **Stronger** ionic bonding → More energy required to break the bonding

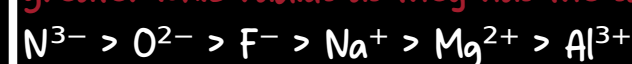
The strength of an ionic bond relies on:

—Ionic radius

Decreasing the ionic radius of the ions in a compound reduces the distance between them which **strengthens** (more exothermic) the forces of attraction (factors affecting the ionic radius were explained in more depth in topic 2 notes)

Example: LiF has a higher lattice enthalpy than KF.

Note: for isoelectronic ions the lower the nuclear charge (number of protons) the greater the ionic radius as they have the same electron configuration and shielding



—Ionic charge

Increasing the charge of ions in the compound **strengthens** (more exothermic) the forces of attraction.

Example: MgO (Mg^{2+} & O^{2-}) has a much higher lattice enthalpy than NaCl (Na^+ & Cl^-).

Ionic bonding

Polarisation

Polarisation: the **distortion** of the **electron density** (electron cloud) of an anion by a cation so that it's no longer spherical.

When the lattice enthalpy is calculated **theoretically** the ions are assumed to be **perfectly spherical**. However, when it is measured (e.g. using Born-Haber cycle) the energy required is greater (more negative) as the bonds are not 100% ionic (it has some covalent characteristics) caused by the polarisation the cation to the anion

Cation

The polarising power (ability to distort) of a cation depends on:

-Ionic radius

A **smaller** cation is **more polarising** than larger one as it makes the outer shell of anion closer to the cation's nucleus.

-Charge

The **greater the charge** of the cation the **stronger** it attracts the outer electrons (more polarising).

The polarisability (ease of polarising) of an anion depends on:

Anion

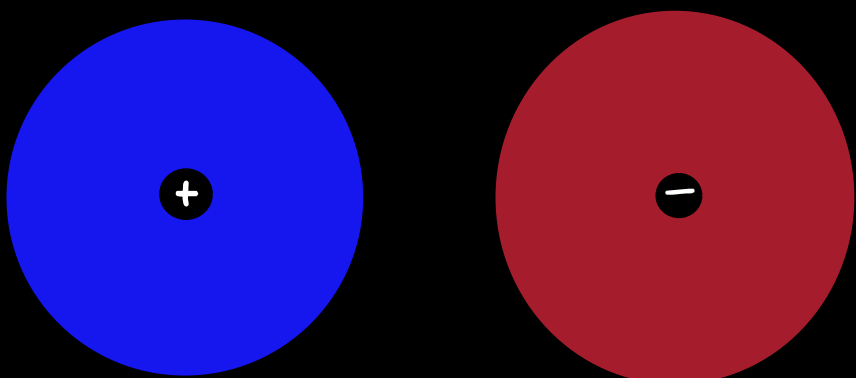
-Ionic radius

A **larger** anion is **more polarisable** than smaller one as the electron cloud would be further from its nucleus so **easier to distort**.

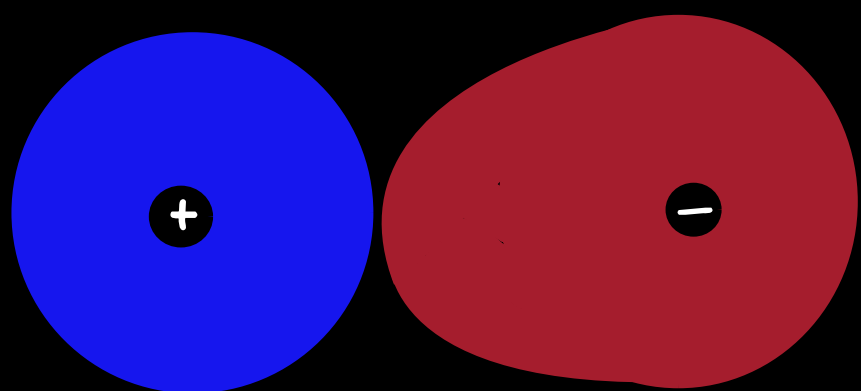
-Charge

The **greater the charge** of the anion the **more polarisable** it would be as there would be **more electron to distort**.

100% Ionic



Ionic with some covalent characteristics



The polarisation creates some degree of electron sharing (covalent)

Covalent bonding

Covalent bond: the **strong** electrostatic attraction between the **two nuclei** of bonded atoms and the **shared pair of electrons**.

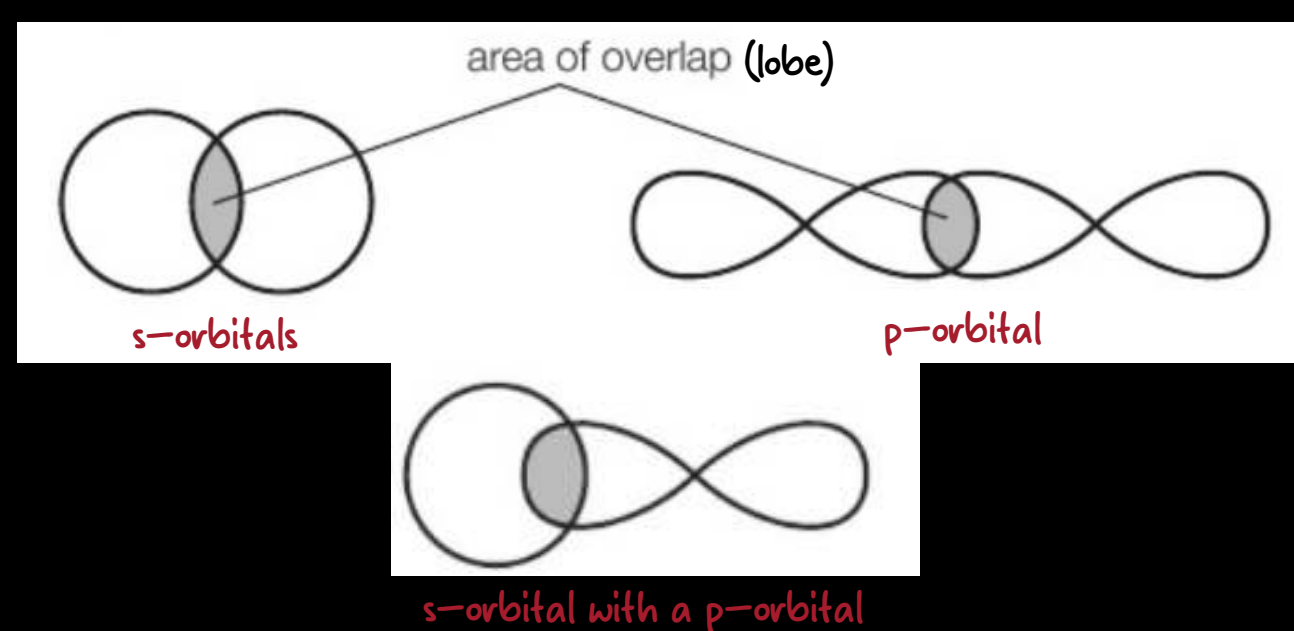
A covalent bond is formed when **two atomic orbitals** in the two atoms containing an **unpaired** (single) electron overlap forming a **molecular orbital** containing paired (two) electrons.

Types of covalent bonds

-Sigma (σ) bond

A sigma bond is formed by an **end-to-end** overlap of atomic orbitals, and the electron is concentrated **between** the nuclei of the bonding atoms along the internuclear axis.

Note: the electron density is symmetrical about the line joining nuclei (internuclear).



-Pi (π) bond

A pi bond is formed by a **side-to-side** overlap of atomic orbitals, and the electron is concentrated **above and below** the plane of the nuclei of the bonding atoms.

Pi bonds are only formed after a sigma bond has been formed, so they are only found in double and triple bonds.

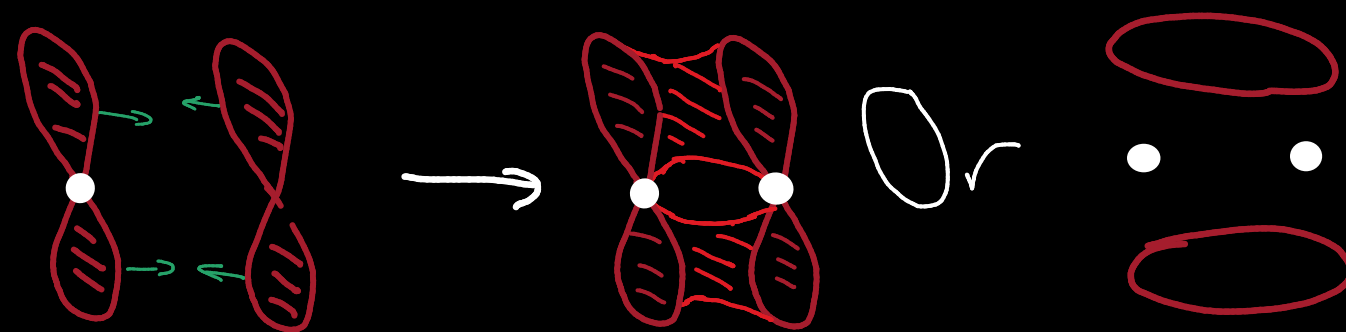
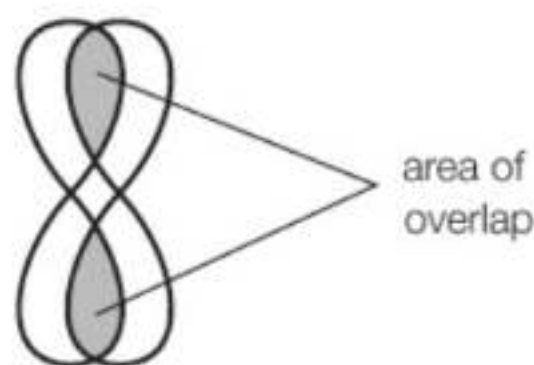
single $\rightarrow$ 1 sigma

double $\rightarrow$ 1 sigma + 1 pi

triple $\rightarrow$ 1 sigma + 2 pi

Note: a pi bond is weaker than a sigma bond. That is why the reactivity of alkenes > alkanes

Reason: sigma forms an end-to-end overlap which is harder to break compared to side-to-side (pi).



This is an acceptable way of writing the molecular orbital forming π bond

Covalent bonding

Bond length and strength

Bond length: The **distance** between two atoms that are covalently bonded together.

The strength of covalent bond relies on the amount of energy required to break 1 mole of the bond in the gaseous state.

—**Relation:**

Generally the shorter the bond length the stronger the attraction hence bond strength.

BOND	BOND LENGTH / nm	BOND STRENGTH / kJ mol ⁻¹
Cl-Cl	0.199	242
Br-Br	0.228	193
I-I	0.267	151
C-C	0.154	347
C=C	0.134	612
C≡C	0.120	838
N-N	0.145	158
N=N	0.120	410
N≡N	0.110	945
O-O	0.148	144
O=O	0.121	498

Fluorine **exception**:

F-F bond is shorter than Cl-Cl but weaker because:

Small atomic size → The small size forces the lone pairs on the two fluorine atoms **extremely close** together creating **strong repulsion** between them, which weakens the bonding interaction.

Note: when comparing bond strength make sure bonds are of similar nature

Electronegativity (E.N.)

Electronegativity: the ability of an atom to attract a bonding pair of electrons in a covalent bond.

H 2.1 Li 1.0 Be 1.5 Na 0.9 Mg 1.2 K 0.8 Ca 1.0 Sc 1.3 Ti 1.5 V 1.6 Cr 1.6 Mn 1.5 Fe 1.8 Co 1.8 Ni 1.8 Cu 1.9 Zn 1.6 Ga 1.6 Ge 1.8 As 2.0 Se 2.4 Br 2.8 Kr Rb 0.8 Sr 1.0 Y 1.2 Zr 1.4 Nb 1.6 Mo 1.8 Tc 1.9 Ru 2.2 Rh 2.2 Pd 2.2 Ag 1.9 Cd 1.7 In 1.7 Sn 1.8 Sb 1.9 Te 2.1 I 2.5 Xe Cs 0.7 Ba 0.9 La 1.1 Hf 1.3 Ta 1.5 W 1.7 Re 1.9 Os 2.2 Ir 2.2 Pt 2.2 Au 2.4 Hg 1.9 Tl 1.8 Pb 1.8 Bi 1.9 Po 2.0 At 2.2 Rn B 2.0 C 2.5 N 3.0 O 3.5 F 4.0 Ne Al 1.5 Si 1.8 P 2.1 S 2.5 Cl 3.0 Ar Sb 1.9 Te 2.1 I 2.5 Xe most E.N. Electronegativity values

observation: non-metals have higher E.N. values.
reason: they have more tendency to gain electrons

observation: noble gases have no E.N. values.
reason: They are unreactive (do not lose nor gain electrons) and would rarely form covalent bonds

Factors affecting electronegativity (attraction):

—**Proton number**

More protons → more attraction

—**Shielding by inner electrons**

More shielding → less attraction

—**Distance from nucleus**

Greater distance → less attraction

Trends:

—**Across the period**

More protons + less distance (atomic radius) + same shell (same shielding)

→ **increase** in E.N. value (more attraction)

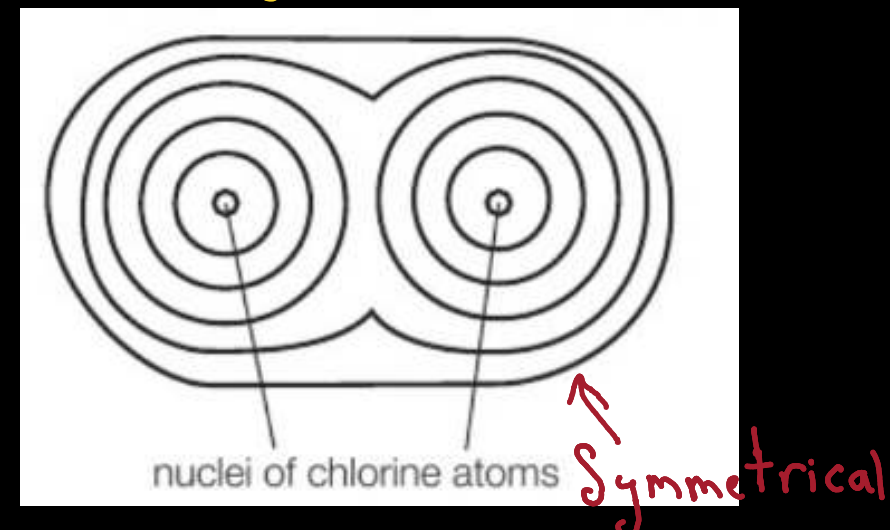
—**Down the group**

Greater distance (atomic radius) + more shells/shielding → **decrease** in E.N. value (less attraction for bonding electrons)

Non-polar (pure) covalent bonds

If two atoms of the same element are bonded the distribution of electrons between the nuclei will be **symmetrical** this is because they have the same E.N. value so their attraction ability is the same (100% covalent)

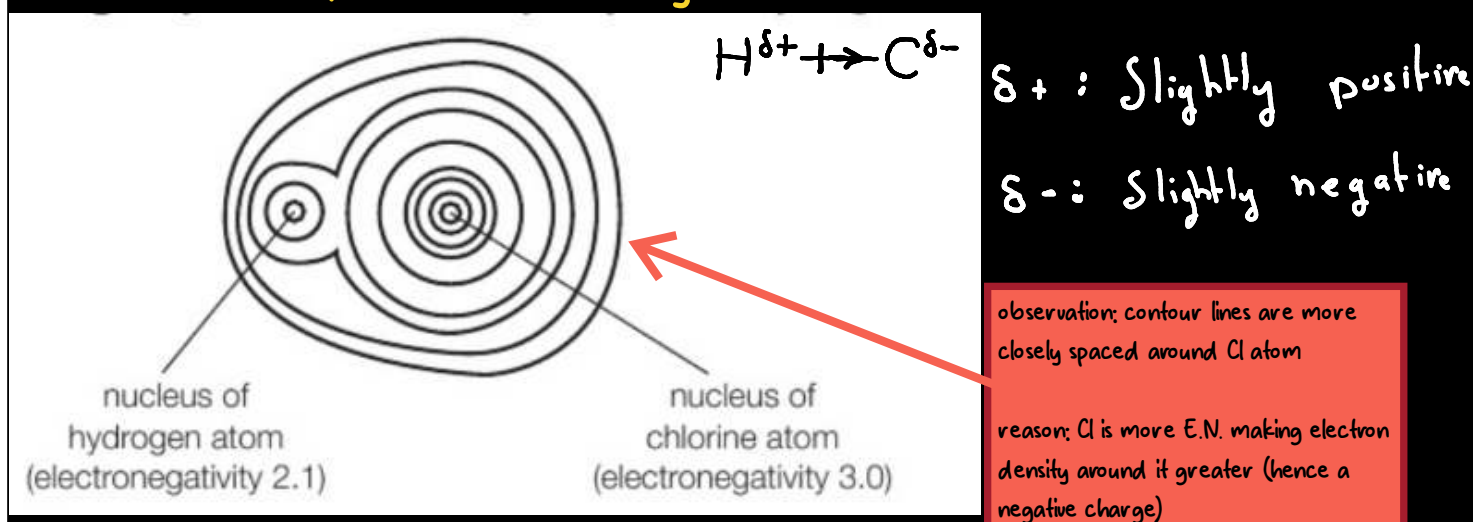
—**Distribution of electron density**



Polar covalent bonds

If there is a difference in E.N. values the distribution of electron density won't be **symmetrical** (unequal distribution). This means there is a separation in charge between two atoms so carries a **slightly positive**(+) charge while the other carries a **slightly negative**(-) charge. (this separation in charge is called a **dipole**)

—**Distribution of electron density**



Continuum of bonding types

In reality, chemical bonding is not purely ionic or covalent. Instead, there is a spectrum, where bonds have varying degrees of ionic and covalent character.

The difference in electronegativity determines the character degree.

ELECTRONEGATIVITY DIFFERENCE	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1
APPROXIMATE % IONIC CHARACTER	0	0.5	1	2	4	6	9	12	15	19	22	26

ELECTRONEGATIVITY DIFFERENCE	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3
APPROXIMATE % IONIC CHARACTER	30	34	39	43	47	51	55	59	63	67	70	74

ELECTRONEGATIVITY DIFFERENCE	2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3
APPROXIMATE % IONIC CHARACTER	76	79	82	84	86	88	89	90	91	92

—**Nonpolar Covalent Bond**: Electronegativity difference is approximately **0** to **0.4**.

—**Polar Covalent Bond**: Electronegativity difference is **greater** than **0.4** but **less** than approximately **1.7–2.0**.

—**Ionic Bond**: Electronegativity difference is **greater** than approximately **1.7–2.0**.

Covalent bonding

Drawing (simple) Covalent bonds

Cross and dot diagram

Octet rule: the outer shell electrons in each atoms must be the same as outer shell electrons in a noble gas

Drawing tips:

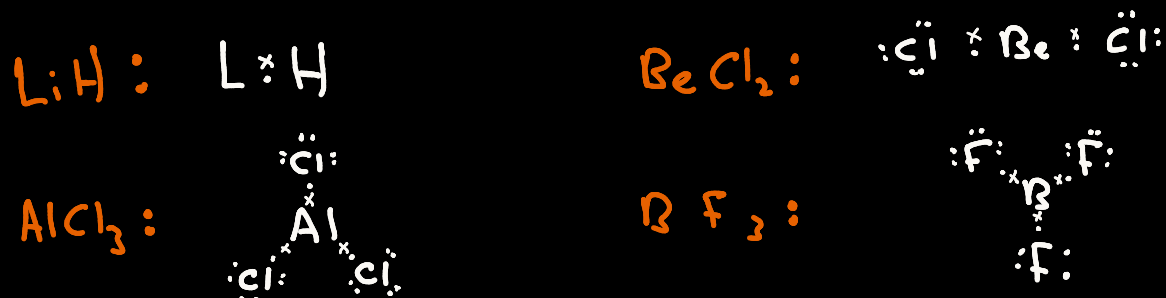
1. The atom with the least electronegativity will be the central atom (exception: hydrogen)
2. Make sure that the total number of electrons stays the same

Examples:

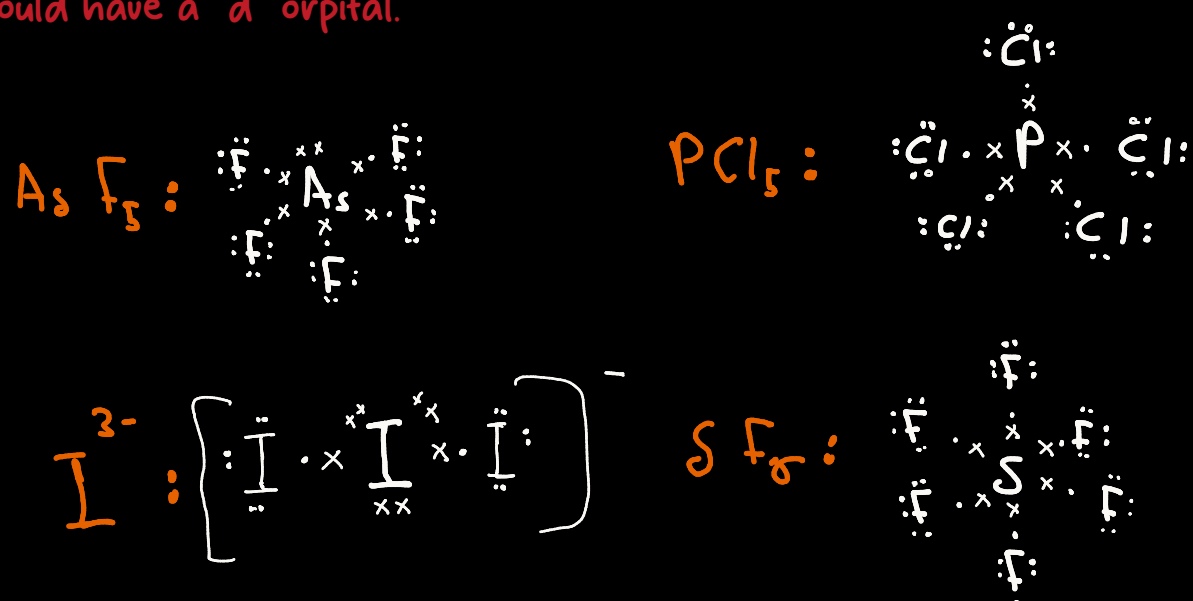


Exceptions to octet rule

Octet Deficient:



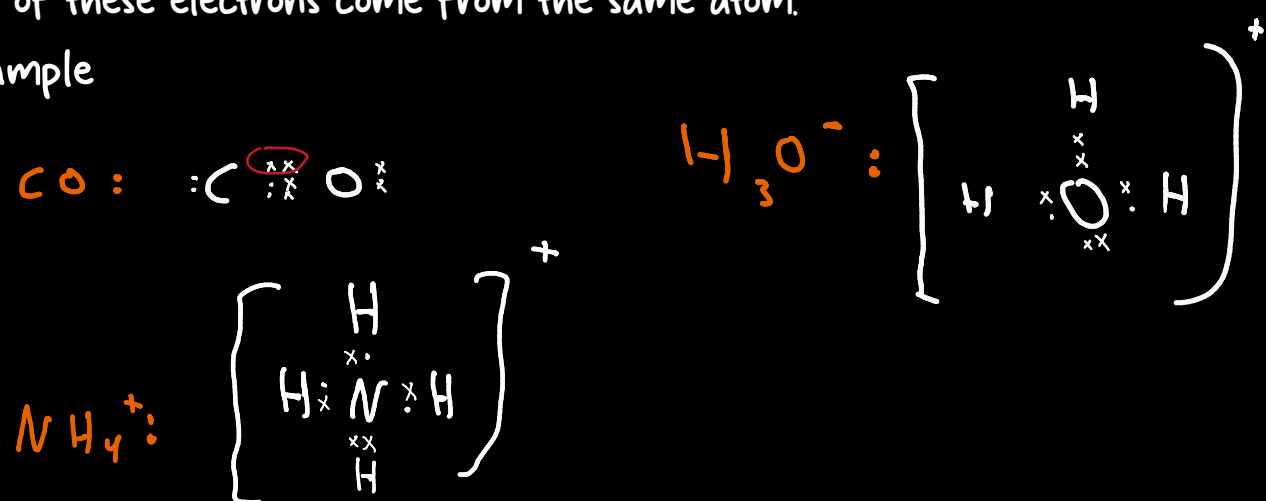
Expanded octet:



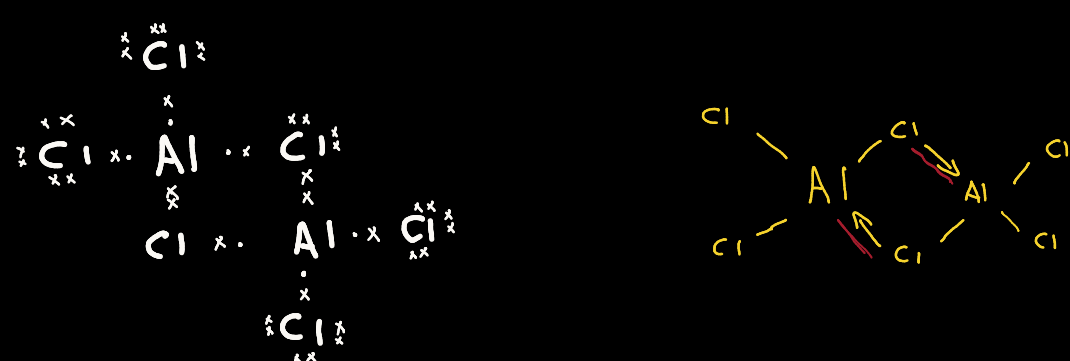
Dative molecules

Dative covalent bond: a pair of electrons shared between two atoms where both of these electrons come from the same atom.

Example



Dimer: two identical molecules joined/held by dative bonds



Shapes of simple molecules and ions

VSEPR (valence shell electron pair repulsion)/ERP theory

This theory states:

The shape of a molecule or ion is caused by **repulsion** between the pairs of electrons, both bonding and lone(non-bonding) pairs that surround the central atom

The electron pairs arrange themselves around the central atom so that repulsions between them is at its **minimum**.

Lone pair-lone pair repulsion > lone pair-bonding pair repulsion > bond pair-bond pair repulsion

How to explain the shape of the molecule/ion?

- 1-State number of sets bonding pairs (double bonds are considered one) and lone pairs
- 2-"They arrange to maximum separation and minimum repulsion"
- 3-

if there are only bonding pairs $\rightarrow$ "pairs repel equally"

if there is only 1 lone pairs and rest are bonding $\rightarrow$ "lone pair-bonding pair repulsion $>$ bond pair-bond pair repulsion"

if there are more than 1 lone pairs and rest are bonding $\rightarrow$ "lone pair-lone pair repulsion $>$ lone pair pair-bond pair repulsion"

- 4-State the angle and shape

"Tetrahedral bond angle (109.5) is reduced" in case of U-shaped and pyramidal

NUMBER OF BOND PAIRS	NUMBER OF LONE PAIRS	SHAPE	EXAMPLE
2	0	linear	
3	0	trigonal planar	
4	0	tetrahedral	
5	0	trigonal bipyramidal	
6	0	octahedral	
3	1	trigonal pyramidal	
2	2	V-shaped	

Covalent bonding

Polar and non-polar molecules

Polar Molecules

Electron Density: Uneven (non-symmetrical). Electrons are pulled closer to the more electronegative atoms, creating a "lopsided" molecule.

Polar Bonds: Must have polar bonds (e.g. O—H, N—H, C—F).

Types of bonds: All bonds can be the same type OR they can have different types (which often contributes to the polarity such as CHCl_3)

Dipole Cancellation: The dipoles from these bonds DO NOT cancel out due to a non-symmetric shape.

Net Dipole Moment: YES. The molecule has a positive end and a negative end.

Example: Water (H_2O) — The bent shape means the two O—H bond dipoles reinforce each other instead of canceling

Non-Polar Molecules

Electron Density: Even (symmetrical). The charge is balanced across the molecule.

Polar Bonds: Can have no polar bonds (e.g. Cl_2 , H_2) OR can have polar bonds.

Types of bonds: If polar bonds are present, they are typically the same type and arranged symmetrically. (It is very rare to have different bond types and still be non-polar)

Dipole Cancellation: If polar bonds are present, their dipoles DO cancel out due to a highly symmetric shape.

Net Dipole Moment: NO. The molecule has no distinct charged ends.

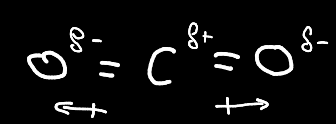
Example: Carbon Dioxide (CO_2) — The linear shape causes the two polar C=O bond dipoles to point in opposite directions and cancel each other out.

Symmetrical and non symmetrical shapes

Symmetrical:

- 1.Linear
- 2.Trigonal planar
- 3.Tetrahedral
- 4.Trigonal bipyramidal
- 5.Octahedral

Example



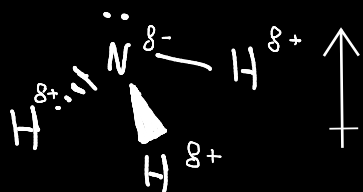
How to explain molecules polarity:

- 1—State the the shape and whether it is symmetrical or not
- 2—The type of bonds whether they are identical/same or not
- 3—Determine Dipole Cancellation
- 4—State net dipole moment

Non-symmetrical:

- 1.Trigonal pyramidal
- 2.U-shaped

Example



Molecular structures

Properties

They have **low melting and boiling** points because the individual molecules are held (within a crystal lattice in solid state) by only **weak intermolecular forces**, not by strong covalent or ionic bonds. Relatively **little energy** is needed to overcome these weak forces and break down the orderly lattice structure, allowing the molecules to escape into the liquid or gas phase.

Note: The strength of these intermolecular forces increases with the size of the electron cloud (more electrons) and the length of the molecule.

Giant Covalent structures

General properties

High Melting and Boiling Points

Explanation: Giant covalent structures have very high melting and boiling points because a **large amount of energy** is required to break the **strong, extensive(many)** network of **covalent** bonds throughout the **giant covalent lattice** structure

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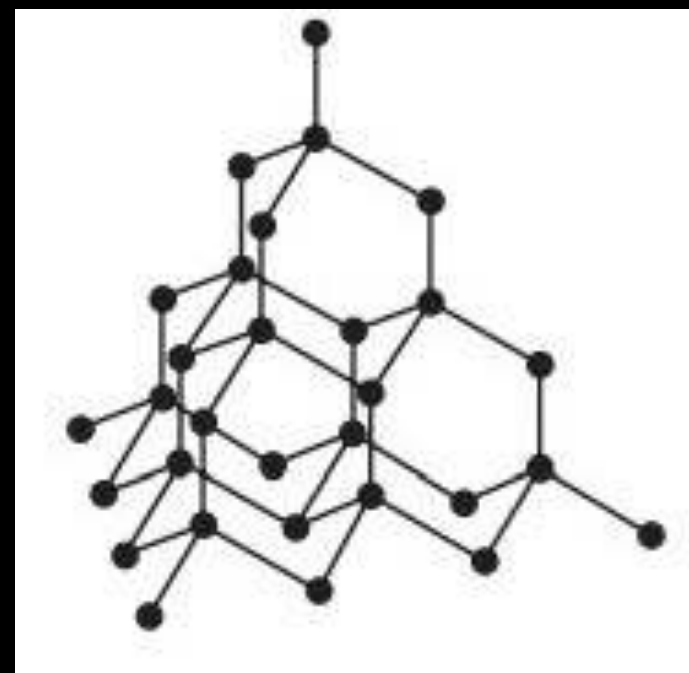
Not Conductive (except graphite/graphene)

Explanation: The outer electrons are fixed in place so there wouldn't be any mobile nor delocalized electrons

Diamond

Description: Each carbon atom is bonded to 4 other carbon atoms (sigma bonds) forming a 3D tetrahedral.

note: all of its bonds are strong covalent bonds with not intermolecular forces making it rigid and strong.



Properties and uses

- 1—One of the hardest substances known (use: drilling and cutting)
- 2—It is a poor conductor of electricity (all electrons are shared hence no mobile delocalized electrons)
- 4—Color from Trace Impurities (use: creates rare and valuable colored gemstones for jewelry)
- 3—High Optical Dispersion ("Fire") (use: creates the sparkle in colorless gemstones for jewelry)

Silica/ Silicon dioxide (SiO_2)

Description: Is very similar to diamond where each silicon atom is bonded to 4 oxygens atoms atoms (sigma bonds) and each oxygen in bonded to 2 silicon atoms forming a 3D tetrahedral structure around Si.

note: SiO_2 represent the ratio of Si:O not the bonding to it must not be mistaken for simple compounds like CO_2

Silicon dioxide vs Diamond

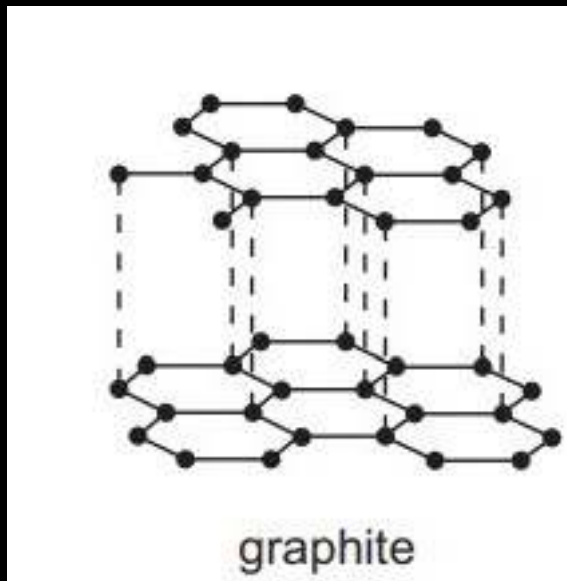
Diamond is a higher melting/ boiling point as there are greater bonds per atom + oxygen(in SiO_2) has lone pairs

Giant Covalent structures

Graphite

Description: Each carbon atom is bonded to 3 other carbon atoms (sigma bonds) forming a layers of hexagonal rings. the layers of hexagonal rings are held by weak intermolecular forces so the layers can slide over each other when a force is applied (soft and malleable)

note: the fourth electron that is not shared becomes delocalized and free to move



Properties and uses

1—Layers can slide over one another/soft dues to weak forces between graphite layers (use: lubricant)

2—Conduct electricity due to mobile delocalized electrons (use: electrodes)
note: unlike metal graphite only conducts in primarily within the planes. The delocalized electrons can move freely across the 2D plane of the hexagonal sheets, but they cannot easily jump between the sheets due to the large separation and weak forces.

3—High melting point due to strong covalent forces within each layer

Graphene

Description: it is a single layer of graphite where each carbon atom is bonded to 3 other carbon atoms (sigma bonds)

note: the fourth electron that is not shared becomes delocalized and free to move.

Properties and uses

1—Electrical Conductivity and transparent (use: transparent electrodes scores)

2—Uncreactive (use: oxidation resistant layer)

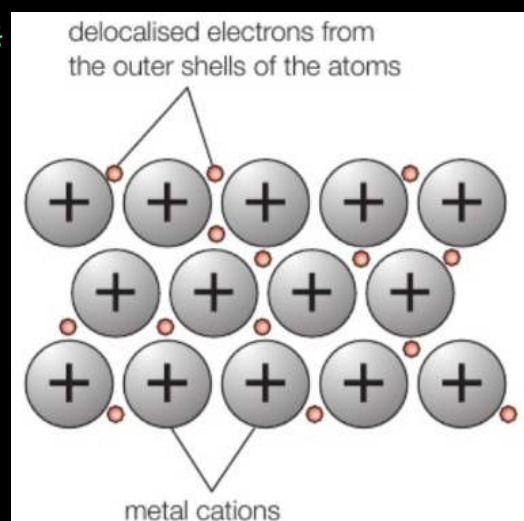
3—Transparent & Conductive (use: flexible electronic such as Touchscreens, Solar Cells)

4— Ultra—Low Friction / Superlubricity (use: low friction coating)

Metallic bonding

Metallic bond: the **strong** electrostatic attraction between the positive metal ions and the sea delocalized electrons surrounding them.

if asked about a specific ion describe the:
1) Charge of cation
2) Num. of electrons per ion



Description: Metals have characteristic properties due to their giant metallic lattice structure, which consists of a regular arrangement of positive ions (**cations**) surrounded by a 'sea' of **delocalized electrons** electrostatically attracted.

Properties

1. High Melting and Boiling Points

Explanation: large amount of energy is required to overcome the **strong** electrostatic forces of attraction (metallic bonds) between the positive metal ions and the sea of delocalized electrons.

2. Good Conductors of Electricity

Explanation: The **delocalized electrons** are free to move(mobile) throughout the metallic lattice. When a voltage (potential difference) is applied across a metal, these mobile electrons **flow**/move forming an electric current.

3. Good Conductors of Heat

Explanation: Heat is transferred efficiently through metals via two main mechanisms:

Electron Movement. The mobile delocalized electrons gain kinetic energy where the metal is heated. They move rapidly throughout the lattice, colliding with ions and other electrons, thereby transferring kinetic (thermal) energy.

Vibrating Ions The positive ions themselves vibrate in fixed positions. As they vibrate more vigorously, the cations transfer energy to their neighbors through collisions.

4. Malleable and Ductile

Malleable: Can be hammered or rolled into sheets.

Ductile: Can be drawn into wires.

Explanation: The layers of positive ions in the metallic lattice **can slide over one another** without shattering the structure. The delocalized **electrons act as a "glue"**(hold together) that maintains the metallic bonding by moving to accommodate the shift. This prevents layers of like—charged ions from coming directly opposite each other and repelling, allowing the metal to be reshaped permanently.

Strength

Factors affecting the strength of metallic bonding (attraction between cation and delocalized electrons):

1—Number of delocalized electrons (**more** electrons → **stronger** bond)

2—Charge (**greater** charge → **stronger** bond)

3—Ionic radius (**smaller** ions → **stronger** bond)

	GIANT METALLIC	GIANT IONIC	GIANT COVALENT	MOLECULAR
Particles present	positive ions and delocalised electrons	positive and negative ions	atoms	molecules
Type of bonding	metallic	ionic	covalent	covalent
Are there any intermolecular forces of attraction?	no	no	no	yes
Melting and boiling temperatures	fairly high to high	fairly high to high	high to very high	generally low
Electrical conductivity	good when solid and when molten	non-conductor when solid; good when molten	non-conductor	non-conductor
Solubility in water	insoluble unless the metal reacts with water, e.g. sodium	generally soluble, but with notable exceptions, e.g. AgCl, AgBr, AgI and BaSO ₄	insoluble	generally insoluble, but may dissolve if hydrogen bonding is possible (e.g. sucrose), or if the substance reacts with water (e.g. Cl ₂) (See Topic 7 for an explanation of hydrogen bonding.)